Spatial–Spectral Attention Graph U-Nets for Hyperspectral Image Classification

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***Abstract*— Graph neural networks (GNNs) have outstanding performance in modeling global spatial information dependence, which makes them very suitable for the complex and diverse distribution of objects in hyperspectral images (HSIs). In recent years, they have been widely used in research on HSI classification. However, the high dimensionality and high redundancy between bands of HSIs make it difficult to explore deep spectral features, resulting in high computational and storage costs for GNNs. In addition, traditional convolutional neural networks (CNNs) preprocess images to obtain pixel-level features, but the local receptive field of their convolutional kernels cannot model nonlocal spatial dependencies, leading to extracted features that lack diversity. To address these issues, this article proposes a graph convolutional network that integrates spatial and spectral attention using the graph U-Nets architecture for HSI classification. First, a spectral pixel sequence (SPS) extraction module is constructed in the encoder stage based on spectral attention to reduce the influence of redundant bands and enable graph convolution to extract more discriminative spectral feature representations. Simultaneously, a lightweight graph attention network (LwGAT) is designed to reduce computing resource consumption during similarity matrix calculation. Second, the multiscale feature aggregation module (MFAM) based on spatial attention is used to extract pixel-level features from the original hyperspectral data, which provides a more diverse spatial–spectral feature for the graph U-Nets by fusing spatial contextual information at different scales. Finally, experiments on four widely used HSI datasets demonstrate that the proposed method achieves better classification performance than other advanced classification methods.**

***Index Terms*— Attention mechanism, graph attention network (GAT), graph U-Nets, hyperspectral image (HSI) classification.**

I. Introduction

With the development of spectral imaging technology, hyperspectral images (HSIs) have more and more spectral bands and spatial texture details, as well as image and spectrum are unified [1]. Hyperspectral remote sensing technology has been widely applied in various fields, such as weather prediction, agricultural detection, soil and water monitoring, and environmental investigation [2], [3], [4], [5], [6]. Hyperspectral image classification is a common processing technology for HSIs. Its main task is to determine the class label for each pixel by utilizing the rich spectral band information and spatial texture structure information of HSIs, thus enabling fine-grained analysis of the observation target area.

In the early stages of remote sensing image classification, traditional classification methods, such as support vector machine (SVM) [7] and random forest [8], were commonly used. To better address the nonlinear classification problems in remote sensing images, kernel skills [9] were introduced to classifier design. This method can effectively transform nonlinear problems in high-dimensional feature space into linear problems in low-dimensional feature space, thereby improving the accuracy and robustness of classifiers. However, classical machine learning algorithms require a heavy and challenging feature engineering stage. In recent years, deep neural networks have made great breakthroughs in nonlinear feature learning. By inputting data into the neural network, the network can automatically implement the feature extraction process from low level to high level, avoiding the tedious manual feature engineering step. Among them, convolutional neural networks (CNNs) have become one of the hot topics in hyperspectral image classification [10] due to its advantages of having fewer parameters, shift invariance, and multiple convolution kernels. To avoid the feature selection being affected by subjective factors, Makantasis et al. [11], Hu et al. [12], Li et al. [13], and Roy et al. [14] used three different CNNs, 1D-CNNs, 2D-CNNs, and 3D-CNNs to encode the spectral, spatial, and spectral–spatial information of pixels, respectively. Zhong et al. [15] introduced residual network into hyperspectral image classification tasks, making the network deeper and improving the network's representation ability. In [16], a lightweight spectral–spatial convolution module was proposed, which greatly reduced the parameters and computational complexity of CNNs. In [17], spatial attention modules and band attention modules were added to the CNNs to reduce the impact of redundant bands and interfering pixels on feature extraction. However, CNNs are difficult to represent the dependence relationship between remote pixels in images due to their fixed convolutional kernel structure, and limited training samples also make CNNs prone to overfitting.

In recent times, researchers have been placing growing emphasis on methods related to graphs [18], [19]. Among them, graph neural networks (GNNs) can globally model any form of graph-structured data and thus have a good performance in characterizing long-range dependencies between

Manuscript received 2 September 2023; revised 25 September 2023; accepted 8 October 2023. Date of publication 17 October 2023; date of current version 26 October 2023. This work was supported in part by the Natural Science Foundation of Ningxia Province of China under Project 2021AAC03179; and in part by the Science Foundation of North Minzu University, Yingchuan, China, under Project 2020KYQD50. *(Corresponding author: Kewen Qu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3324977

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data. GNNs have been widely used in various fields, such as image segmentation [20], text classification [21], and autonomous driving [22], and have also received widespread attention in hyperspectral image classification. For example, Mou et al. [23] proposed a nonlocal graph convolutional network classification method, which simultaneously trains the network using both labeled and unlabeled samples to solve the overfitting problem caused by scarce labeled samples. However, when stacking too many layers, GNNs are prone to oversmoothing [24], resulting in inaccurate representations of extracted nodes. To solve this problem, Bai et al. [25] incorporated dilated convolutions [26] and dense connections [27] to the graph convolutional model. Yang et al. [28] proposed a method to build dynamic graphs across different levels, able to adaptively integrate the initial graph structure. They also used residual mapping and identity mapping to significantly increase the depth of the network, thereby obtaining more abstract feature representation while addressing the oversmoothing problem. In addition, to reduce memory consumption when modeling the graph structure, Hong et al. [29] adopted the method of selecting partial pixels each time to construct the local graph structure. However, the limited number of pixel nodes also limits the expression ability of GNNs. Wan et al. [30] proposed using superpixels instead of pixels for global modeling to reduce memory consumption. However, superpixel segmentation may cause the original pixels of different categories to be assigned to the same superpixel region. In order to solve the problem of incorrectly divided pixels, Bai et al. [31] designed a superpixel feature subdivision mechanism (SFS). This mechanism assigns the extracted superpixel features to the original pixels with different weights. However, the method of modeling based on superpixels cannot capture spectral–spatial features at the pixel level. Thus, Liu et al. [32] combined CNNs and graph convolution to generate complementary spectral–spatial features at both pixel and superpixel levels. To better utilize spectral information, Dong et al. [33] introduced the channel attention module into the CNNs branch based on Liu et al. [32]. Recently, Yu et al. [34] sought to maximize the potential of CNNs and graph convolution; in the graph convolution branch, they randomly eliminated a subset of edges from the input graph while introducing residual connections. This approach effectively mitigates the problem of oversmoothing. Simultaneously, in the CNNs branch, they devised a hybrid attention mechanism to extract spectral–spatial features. Since the above methods based on superpixel segmentation all adopt superpixel structure with fixed proportion segmentation, they cannot learn representations of different spatial topologies of objects. Liu et al. [35] divided hyperspectral images into superpixels of different scales to learn multiscale superpixel features.

However, the graph convolution-based methods mentioned above ignore the generalization of spectral features for hyperspectral image interpretation. This is because graph convolution essentially relies on Laplacian smoothing to make the spectral features of similar pixels similar and enhance the distinguishability of spectral features of different pixels. Therefore, it is worth thinking about how to reduce the influence of redundant bands on spectral feature extraction while using graph convolution to model spatial relationships, so as to obtain more discriminative spectral features through graph convolution aggregation. Meanwhile, the above methods of using CNNs instead of graph convolution to extract pixel-level features solve the huge memory consumption when using pixels as nodes to model the graph structure. However, the limited receptive field of CNNs restricts their ability to model contextual information, thereby making it difficult to establish nonlocal dependencies. This limitation leads to a lack of diversity in the extracted pixel-level features, which cannot meet the requirements of different datasets.

To address the aforementioned issues, this article proposes a spatial–spectral attention graph U-Nets model, which is called SSAGU. First, the spectral pixel sequence (SPS) extraction module was constructed in the graph U-Nets based on multiscale superpixel segmentation. This module can generate SPS representations adapted to superpixels of different scales based on the corresponding superpixel structure graphs. By combining the extracted SPS with graph convolution, the graph convolution can focus on extracting spectral features with intraspectral consistency. Second, we introduce a multiscale feature aggregation module (MFAM) for extracting pixel-level features. By fusing spatial contextual information of different scales, the module further builds nonlocal spatial dependencies to refine the extracted pixel-level spatial–spectral features. Finally, in order to reduce the memory consumption of graph convolution, a lightweight graph attention network (LwGAT) was developed to model the global graph structure.

The main contributions of this article are summarized as follows.

1) In this article, we propose a novel spatial–spectral attention graph U-Nets model, which is called SSAGU. An SPS extraction module is designed in the graph U-Nets architecture to obtain discriminative SPS for identifying land cover classes. By combining graph convolution with SPSs, the reliability of graph convolution's spatial prior is ensured while reducing the influence of redundant spectral bands and making graph convolution more focused on spectral feature extraction.
2) The MFAM is integrated into CNNs for pixel-level feature extraction and reconstruction. MFAM integrates multiple spatial context information to simulate the nonlocally dependent modeling capability of graph convolution models. This integration results in a more diverse pixel-level feature set.
3) An LwGAT is proposed. By discarding the redundant computing parts in graph attention network (GAT), its calculation is simplified, and the computation complexity of the model is reduced while maintaining its accuracy, leading to improved training speed.

The organization of this article is given as follows. Section II briefly reviews related work, including graph U-Nets, superpixel segmentation, and static attention. Section III describes the proposed graph U-Nets model and the proposed spatial–spectral attention strategy, LwGAT, MFAM, and so on. Section IV is devoted to experiments and result analysis, including algorithm comparison, experimental

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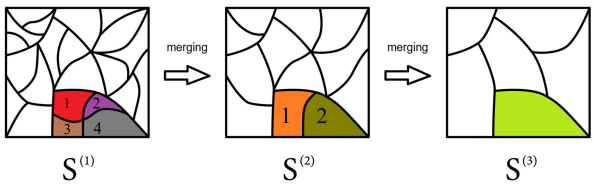


Fig. 1. Schematic of the multiscale superpixel segmentation.

setup, parameter analysis, and ablation experiments. Finally, conclusions of our work are presented in Section V.

II. Related Works

A. Graph U-Nets and Multiscale Superpixel Segmentation

Although architectures of codecs such as U-Nets have been effectively applied to various pixel-level image tasks [36], [37], [38], there is currently a notable absence of analogical codec architectures for graphical data, mainly because pooling and upsampling techniques are not natural operations for this type of data. Gao and Ji [39] devised graph-based graph pooling (gPool) and graph unpooling (gUnpool) operations, which facilitated the integration of graph convolution into U-Nets and effectively countered the problems of oversmoothing related to graph convolution. Due to its novelty, the technique is named graph U-Nets.

Liu et al. [35] utilized multiscale superpixel segmentation along with graph U-Nets to improve hyperspectral image classification. Initially, by implementing the Boruvka algorithm [40], the researchers segmented the original image into superpixels of diverse scales. Afterward, they modeled the graph structure of each graph U-Nets layer by utilizing different scales of superpixels. Finally, the gPool and gUnpool operations based on superpixels were redefined by the researchers to integrate the various scales of superpixels. The process is presented in Fig. 1, where $S^{(l)}$ denotes the superpixel segmentation at layer l. To calculate the value of the superpixels at layer l, we perform gPool by calculating the average of the adjacent superpixels in layer $l-1$, as shown in the following:

$$C_1^{(2)} = \frac{C_1^{(1)}+C_3^{(1)}}{2}, \quad C_2^{(2)} = \frac{C_2^{(1)}+C_4^{(1)}}{2} \tag{1}$$

where $C_u^{(l)}$ denotes the uth superpixel in $S^{(l)}$, and this process is commonly known as gPool operations based on superpixels. Liu et al. [35] extracted spatial–spectral features of superpixels at different scales by performing graph convolution operations on each layer of graph U-Nets. To transfer features linearly from one graph to another, superpixel-based gPool and gUnpool operations are employed. The final model is trained using a cross-entropy loss function to correctly identify the land cover category corresponding to each pixel.

The graph U-Nets utilize multiple layers of graph convolution operations, which exacerbate the deleterious effects of redundant spectral information on feature extraction. Moreover, CNNs fail to establish long-range spatial relationships, leading to a shortage of nonlocal spatial information in the pixel-level features extracted. Thus, we propose constructing an SPS extraction module in this article to eliminate spectral redundancy and utilizing MFAM to flexibly construct nonlocal spatial dependencies.



Fig. 2. Schematic of the static attention.

B. Static Attention

Graph convolutional operation is typically categorized into two domains: spectral domain graph convolution and spatial graph convolutional operation. The latter has seen significant progress due to its simple definition and flexible nature [41], [42]. GAT [43] defines graph convolutional operation as the selective aggregation of neighboring nodes utilizing attention mechanisms. This process is accomplished by using self-attention mechanisms for modeling the correlation among the central node and its neighboring nodes

$$e_{i,j} = \text{LeakyReLU}(\boldsymbol{a}^T[\boldsymbol{W}_1\boldsymbol{v}_i \| \boldsymbol{W}_1\boldsymbol{v}_j]) \tag{2}$$

where $\boldsymbol{v}_i \in R^s$ and $\boldsymbol{v}_j \in R^s$ denote the feature vectors of nodes i and j, respectively, and s denotes the dimension of the feature space. To enhance the feature representation, the authors use $\boldsymbol{W}_1 \in R^{\omega\times s}$ to perform a linear transformation on the feature vectors $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$. $\boldsymbol{a} \in R^{2\omega}$ is a single-layer feedforward neural network, and the output $e_{i,j}$ denotes the attention coefficient between nodes i and j. The vector concatenation operation is denoted by $\|$.

However, Brody et al. [44] demonstrated that the computational function of the attention coefficient in GAT is static, meaning that the weights of the neighborhoods are monotonic. As shown in Fig. 2, while not all query nodes $\{q0, \ldots, q5\}$ have the same attention coefficient for the key node $\{k0, \ldots, k5\}$, they all primarily concentrate on the second key $(k2)$. In addition, the attention coefficient for the key nodes is consistently ranked among all query nodes, where a darker color denotes a higher attention factor. Given a graph $G = (V, E)$ used for GAT modeling, V denotes all nodes within it, while E denotes the edges that connect the nodes. The learnable parameter $\boldsymbol{a}$ in (2) can be expressed as $\boldsymbol{a} = [\boldsymbol{a}_0 \| \boldsymbol{a}_1] \in R^{2\omega}$, $\boldsymbol{a}_0, \boldsymbol{a}_1 \in R^{\omega}$. Therefore, (2) can be written as

$$e_{ij} = \text{LeakyReLU}\left(\boldsymbol{a}_0^T\boldsymbol{W}_1\boldsymbol{v}_i + \boldsymbol{a}_1^T\boldsymbol{W}_1\boldsymbol{v}_j\right) \tag{3}$$

as V is a finite set, there exists a node $j_{\max} \in V$ such that

$$\boldsymbol{a}_1^T\boldsymbol{W}_1\boldsymbol{v}_{j_{\max}} > \boldsymbol{a}_1^T\boldsymbol{W}_1\boldsymbol{v}_j \quad (\forall j \in V) \tag{4}$$

add $\boldsymbol{a}_0^T\boldsymbol{W}_1\boldsymbol{v}_i$ to both sides of the formula (4) simultaneously

$$\boldsymbol{a}_0^T\boldsymbol{W}_1\boldsymbol{v}_i + \boldsymbol{a}_1^T\boldsymbol{W}_1\boldsymbol{v}_{j_{\max}} > \boldsymbol{a}_0^T\boldsymbol{W}_1\boldsymbol{v}_i + \boldsymbol{a}_1^T\boldsymbol{W}_1\boldsymbol{v}_j \quad (\forall j \in V) \tag{5}$$

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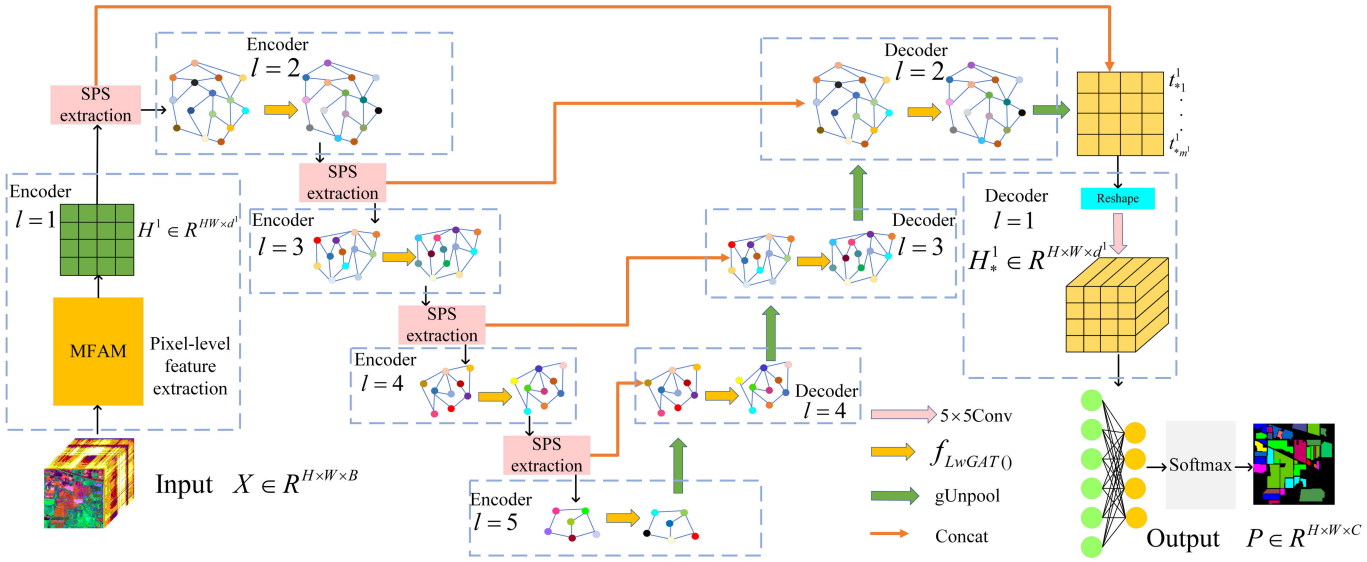


Fig. 3. Overview of the spatial–spectral attention graph U-Nets for HSI classification framework.

and since the LeakyReLU function is monotonic, it can be observed that for $\forall i \in V$, node $j_{\max}$ has the maximum attention coefficient. Therefore, the computational function of the attention coefficient in (2) is static.

III. Proposed Method

This section provides a detailed description of the proposed method, as shown in Fig. 3. For data processing, we first feed the hyperspectral data into MFAM to obtain pixel-level features. We then feed the extracted pixel-level features into graph U-Nets to further extract superpixel-level spatial–spectral information at different scales. In the encoder stage of the graph U-Nets, we utilize LwGAT to model the graph structure for different scales, thereby enabling graph convolution operations. The features produced by the graph convolution are then inputted into the SPS extraction module, whose aim is to adaptively transform features from one graph to another while simultaneously obtaining SPS tailored to different graph structures. In the decoding stage, the SPS extracted by the encoder is used to construct the spectral graph convolution model to extract fine-grained features.

A. Notations

In this work, we define the input hyperspectral image as $\boldsymbol{X} \in R^{H\times W\times B}$, where H, W, and B denote the image's height, width, and number of bands, respectively. We define each hyperspectral pixel as $\boldsymbol{x}_i \in R^B = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$, where n denotes the total number of pixels. We define the result of the Boruvka algorithm segmentation as a multilayer superpixel region, denoted as $s_i^l = \{s_1^l, \ldots, s_{m^l}^l\}$, where l denotes the number of the layers of superpixel segmentation and m^l denotes the number of superpixels in the layer l. When $l = 1$, $m^1 = n$. We define $\boldsymbol{H}^l \in R^{m^l\times d^l} = \{\boldsymbol{h}_1^l, \ldots, \boldsymbol{h}_{m^l}^l\}$ and $\boldsymbol{H}_*^l \in R^{m^l\times d^l} = \{\boldsymbol{h}_{*1}^l, \ldots, \boldsymbol{h}_{*m^l}^l\}$ (particularly, $\boldsymbol{H}_*^1 \in R^{H\times W\times d^1}$) to denote the features extracted during the encoder and decoder stages, respectively, while d^l is the dimension of the feature. Specifically, when $l = 1$, $\boldsymbol{H}^l$ and $\boldsymbol{H}_*^l$ denote the extracted pixel-level features. When l takes on values of 2, 3, 4, or 5, $\boldsymbol{H}^l$ and $\boldsymbol{H}_*^l$ denote the extracted superpixel-level feature at different scales.

B. SPS Extraction Module

In the encoder stage, the input features of layer l are obtained by weighting and merging the output features of layer $l-1$ ($2 \le l \le 5$). It is important to note that this process is performed sequentially, layer by layer. This process is carried out as follows:

$$\boldsymbol{t}_i^l = \mathrm{WM}\left(\boldsymbol{h}_j^{l-1}\right) \quad \text{if } s_j^{l-1} \in s_i^l \quad (1 \le j \le m^{l-1}, 1 \le i \le m^l) \tag{6}$$

the process of weighting and merging WM($\cdot$) is represented by the following implementation steps:

$$Z_{ij}^{l-1} = \boldsymbol{q}^{l-1}\boldsymbol{h}_j^{l-1}, \quad \text{if } s_j^{l-1} \in s_i^l \tag{7}$$

$$Q_{ij}^{l-1} = \mathrm{softmax}\left(Z_{ij}^{l-1}\right) = \frac{\exp(Z_{ij}^{l-1})}{\sum_{s_k^{l-1}\in s_i^l} \exp(Z_{ik}^{l-1})}, \quad (1 \le k \le m^{l-1}) \tag{8}$$

$$\boldsymbol{t}_i^l = \sum_{s_j^{l-1}\in s_i^l} Q_{ij}^{l-1}\boldsymbol{h}_j^{l-1} \tag{9}$$

and the weight extraction vector for the output features of layer $l-1$ is denoted as $\boldsymbol{q}^{l-1} \in R^{d^{l-1}}$. To balance the difference in the contribution of the output features of the current layer to the input features of the next layer and avoid creating extreme weights, the softmax function is utilized to normalize the weights Z_{ij}^{l-1} of the features to Q_{ij}^{l-1}. Then, by weighting and combining the output features of layer $l-1$, the input feature $\boldsymbol{t}_i^l \in R^{d^{l-1}}$ of the node i of layer l is obtained. The objective of this step is to adaptively aggregate the output features of the preceding layer into the input features of the

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next layer in order to mitigate pixel misallocation due to the segmentation of superpixels.

Two processing methods are available for the superpixel feature $\boldsymbol{t}_i^l$ obtained through weighted merging in the previous stage. First, we can use it as an input feature for the next layer of graph convolution. Second, we can input it into a bottleneck block for processing, which is computed as follows:

$$\hat{\boldsymbol{t}}_i^l = \boldsymbol{W}_3^l \partial\left(LN\left(\boldsymbol{W}_2^l\left(\boldsymbol{t}_i^l\right)\right)\right) \tag{10}$$

$$\boldsymbol{H}^l = f_{\mathrm{LwGAT}}(\boldsymbol{t}_1^l, \ldots, \boldsymbol{t}_{m^l}^l), \quad l = 2, 3, 4, 5 \tag{11}$$

where $f_{\mathrm{LwGAT}}(\cdot)$ denotes the LwGAT module, $\hat{\boldsymbol{t}}_i^l$ denotes the band weight coefficient, $\partial(\cdot)$ denotes the relu activation function, and $\mathrm{LN}(\cdot)$ denotes the layer normalization function. We use $\boldsymbol{W}_2^l \in R^{d^{l-1}/\mathrm{r} \times d^{l-1}}$ and $\boldsymbol{W}_3^l \in R^{d^{l-1} \times d^{l-1}/r}$ as trainable parameters of the bottleneck block. The hyperparameter r is manually set and will undergo a detailed analysis within our experimental setup. In order to generate the SPS of layer $l-1$, we conduct addition operations at the elemental level, and these operations involve the band weight coefficient $\hat{\boldsymbol{t}}_i^l$ and the output features from the layer $l-1$. The calculation process is given as follows:

$$\boldsymbol{sps}_j^{l-1} = \hat{\boldsymbol{t}}_i^l \oplus \boldsymbol{h}_j^{l-1}, \quad s_j^{l-1} \in s_i^l \tag{12}$$

where $\oplus$ denotes the elemental-level addition operation and $\boldsymbol{sps}_j^{l-1}$ denotes the SPS of node j of layer $l-1$, which is produced during the encoder stage.

In the decoder stage, we concatenate the SPS, generated in the encoder stage with the superpixel features generated in the decoder stage. This combined sequence serves as the input features for the graph convolution in the decoder stage as follows:

$$\boldsymbol{t}_{*j}^l = \mathrm{Concat}\left(\boldsymbol{sps}_j^l, \boldsymbol{h}_{*i}^{l+1}\right), \quad s_j^l \in s_i^{l+1} \tag{13}$$

where $\mathrm{Concat}(\cdot)$ denotes a concatenation operation, $\boldsymbol{h}_{*i}^{l+1}$ denotes the output features of node i of layer $l+1$ of the decoder, among them, when $l = 5$, $\boldsymbol{H}_*^l = \boldsymbol{H}^l$, and $\boldsymbol{t}_{*j}^l \in R^{d^l + d^{l+1}}$ denotes the input features of node j of layer l of decoder. During the decoder stage, the input features of layer l of the decoder are created by concatenating together the output features of layer $l+1$ of the decoder with the SPS generated in layer l of the encoder. We then input the concatenated features $\boldsymbol{t}_{*j}^l$ into LwGAT to construct the spectral graph convolution module, as shown in the following equation:

$$\boldsymbol{H}_*^l = f_{\mathrm{LwGAT}}\left(\boldsymbol{t}_{*1}^l, \ldots, \boldsymbol{t}_{*m^l}^l\right), \quad l = 2, 3, 4 \tag{14}$$

when $l = 1$

$$\boldsymbol{H}_*^l = \mathrm{Conv}\left(\mathrm{Reshape}\left(\boldsymbol{t}_{*1}^l, \ldots, \boldsymbol{t}_{*m^l}^l\right)\right) \tag{15}$$

$$\boldsymbol{P} = \mathrm{softmax}\left(\mathrm{FCL}\left(\boldsymbol{H}_*^l\right)\right) \tag{16}$$

where $\mathrm{Reshape}(\cdot)$ maps the input features from 2-D to 3-D ($R^{m^1 \times (d^l + d^{l+1})} \rightarrow R^{H \times W \times (d^l + d^{l+1})}$), when $l = 1$, $m^1 = H \times W$, and $\mathrm{Conv}(\cdot)$ denotes the convolution operation. The extracted features are then transformed into probabilistic labels by the fully connected layer $\mathrm{FCL}(\cdot)$. The final classification result $\boldsymbol{P} \in R^{H \times W \times C}$ is obtained using the softmax function, where C denotes the total number of categories.

Through this module, we obtained a nonredundant SPS representation in the encoder stage, providing a more fine-grained input feature for graph convolution in the decoder stage. The specific details are shown in Fig. 4.

C. Lightweight Graph Attention Network

In recent years, attention modules have been widely used to obtain node embeddings in GNNs. The GAT, which uses attention mechanism-based neighborhood aggregation, was introduced in [43] to compute attention coefficients between nodes in a graph, thereby producing a global graph structure. However, Brody et al. [44] demonstrated that the computational function of attention coefficient in traditional GAT is static. According to formulas (4) and (5), we can infer that in GAT, the attention coefficient e_{ij} is only dependent on node j. This implies that the outcome of e_{ij} is determined by $\boldsymbol{a}_1^T \boldsymbol{W}_1 \boldsymbol{v}_j$ and is independent of $\boldsymbol{a}_0^T \boldsymbol{W}_1 \boldsymbol{v}_i$. Therefore, in our LwGAT, we omit the process of calculating $\boldsymbol{a}_0^T \boldsymbol{W}_1 \boldsymbol{v}_i$. The specific details are shown in Fig. 5.

Let us consider the encoder stage as an example. The representation of LwGAT is given as follows:

$$\begin{aligned} \mathrm{GAT}: \widetilde{e}_{ij}^l &= \mathrm{LeakyReLU}\left((\widetilde{\boldsymbol{a}}^l)^T [\boldsymbol{W}^l \boldsymbol{t}_i^l \| \boldsymbol{W}^l \boldsymbol{t}_j^l]\right) \\ &= \mathrm{LeakyReLU}\left(\left(\widetilde{\boldsymbol{a}}_0^{\,l}\right)^T \boldsymbol{W}^l \boldsymbol{t}_i^l + \left(\widetilde{\boldsymbol{a}}_1^{\,l}\right)^T \boldsymbol{W}^l \boldsymbol{t}_j^l\right) \end{aligned} \tag{17}$$

$$\mathrm{LwGAT}: \widetilde{e}_{ij}^l = \mathrm{Sigmoid}\left(\left(\widetilde{\boldsymbol{a}}_1^l\right)^T \boldsymbol{W}^l \boldsymbol{t}_j^l\right) \tag{18}$$

and to mitigate the impact of the monotonicity of the activation function on the ultimate outcome, we opted to utilize the sigmoid activation function in lieu of the original leakyrelu activation function, where $\boldsymbol{W}^l \in R^{\widetilde{\omega}^l \times d^{l-1}}$ denotes a linear transformation of the features $\boldsymbol{t}_i^l$ and $\boldsymbol{t}_j^l$, aimed at enhancing the feature representation. In addition, $\widetilde{\boldsymbol{a}}^l \in R^{2\widetilde{\omega}^l}$, $\widetilde{\boldsymbol{a}}_0^l \in R^{\widetilde{\omega}^l}$, and $\widetilde{\boldsymbol{a}}_1^l \in R^{\widetilde{\omega}^l}$ denote a feedforward neural network. It is evident that the training parameters for the feedforward neural network $\widetilde{\boldsymbol{a}}_1^l$ in LwGAT are half of those for the feedforward neural network $\widetilde{\boldsymbol{a}}^l$ in GAT. As the adjacency matrix constructed by GAT is not symmetric (i.e., $\widetilde{e}_{ij}^l \neq \widetilde{e}^l{}_{ji}$), the computational complexity required for GAT to calculate the correlation between nodes at layer l is $O(|\widetilde{V}^l| \times \widetilde{\omega}^l \times d^{l-1}) + O(|\widetilde{V}^l|^2 \times 2\widetilde{\omega}^l)$. In contrast, the computational complexity required for LwGAT is $O(|\widetilde{V}^l| \times \widetilde{\omega}^l \times d^{l-1}) + O(|\widetilde{V}^l| \times \widetilde{\omega}^l)$. Here, $|\widetilde{V}^l|$ denotes the number of nodes in the layer l. We then use the softmax function to normalize $\widetilde{e}_{ij}^l$ into the similarity coefficient $\widetilde{\alpha}_{ij}^l$

$$\widetilde{\alpha}_{ij}^l = \frac{\exp(\widetilde{e}_{ij}^l)}{\sum_{k \in N_i} \exp(\widetilde{e}_{ik}^l)} \tag{19}$$

where N_i is the superpixel that shares a common edge with superpixel s_i^l (i.e., the neighborhood of node i). To enhance the robustness of the model and prevent overfitting, we incorporated a dropout operation. This operation randomly sets the similarity coefficient of some neighborhood nodes to zero, which prevents the center nodes from relying on specific neighborhood nodes to form dependencies and enhances generalization. Therefore, our graph convolution expression is

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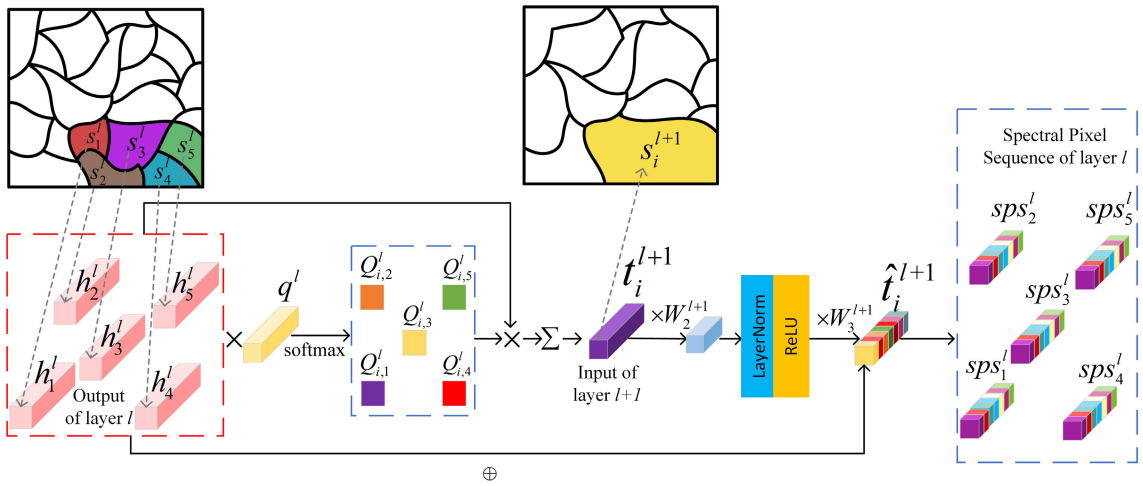


Fig. 4. Structure of the SPS extraction module, $\oplus$ stands for elemental-level addition, and $\sum$ denotes the summation.

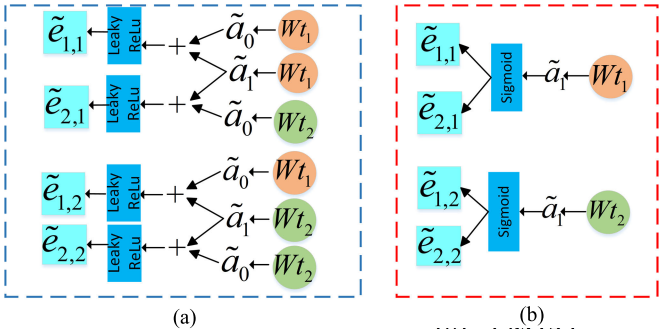


Fig. 5. Difference between (a) GAT and (b) LwGAT in calculating attention coefficient $\widetilde{e}_{ij}^{l}$. Here, we take two nodes as an example, and t_1 and t_2 denote the input features of the two nodes. The GAT model requires the inclusion of the extra training parameter $\widetilde{a}_0$, as well as four additional addition operations. Assuming that there are n nodes in the graph, GAT will perform an additional n^2 addition operations. It is evident that as the number of nodes in the graph increases, the advantage of LwGAT will continue to expand.

given as follows:

$$\boldsymbol{h}_i^l = \sum_{j \in N_i} \text{dropout}\left(\widetilde{\alpha}_{ij}^l\right) \boldsymbol{W}^l \boldsymbol{t}_j^l \tag{20}$$

where $\boldsymbol{h}_i^l$ denotes the output features of node i of layer l. Empirically, we set the dropout to 0.4.

D. Multiscale Feature Aggregation Module

Although graph convolution can model remote dependencies and enhance spatial information, it is better suited for modeling larger structures, such as superpixels. The use of pixel-level nodes to model graph structures can result in significant computational overheads, making it challenging to apply in real-world scenarios. This work employs MFAM [45] as a module for extracting pixel-level features. The capabilities of the module for multiscale contextual interaction are utilized to simulate the nonlocal dependency of graph convolution, facilitating the extraction of pixel-level features with heightened discriminative power, as shown in Fig. 6.

The MFAM comprises two cascade components, namely, FD($\cdot$) and Moga($\cdot$)

$$\boldsymbol{F} = \sigma(\text{Conv}(\text{Norm}(\boldsymbol{X})) + b_{\text{cnn}}) \tag{21}$$

$$\boldsymbol{T} = \text{Moga}(\text{FD}(\boldsymbol{F})) \tag{22}$$

where Conv($\cdot$) and b_{cnn} denote the convolution operation and bias, respectively, and $\sigma(\cdot)$ denotes the leakyrelu activation function. The variable $\boldsymbol{F} \in R^{H\times W\times d^1}$ denotes the data obtained by the CNNs with a constant spatial dimension and compressed spectral dimension. Moga($\cdot$) is a multiorder gated aggregation module that comprises a gated branch (F_ϕ) and a contextual aggregation branch (G_ψ), which produces an output $\boldsymbol{T} \in R^{H\times W\times d^1}$. FD($\cdot$) is designed to decompose features and encourage the network to prioritize multiorder interactions while dynamically excluding trivial interactions. The formula for FD($\cdot$) is given as follows:

$$\boldsymbol{Z} = \text{GELU}(\boldsymbol{F} \oplus \gamma_s \odot (\boldsymbol{F} - \text{GAP}(\boldsymbol{F}))) \tag{23}$$

where $\odot$ denotes the elemental-level multiplication, $\gamma_s \in R^{d^1\times 1}$ is a scale factor initialized to zero, GAP($\cdot$) refers to global average pooling, and FD($\cdot$) enhances the diversity of features by reweighting the trivial interaction component $\boldsymbol{F} - \text{GAP}(\boldsymbol{F})$.

The multiorder gated aggregation module (Moga($\cdot$)) integrates depthwise convolution (DWConv) [46] to encode multiscale features. The module utilizes three DWConv layers to capture low-, medium-, and high-scale interactions in parallel, where the expansion ratio is $e \in \{1, 2, 3\}$. The process is given as follows. First, when the input features $\boldsymbol{Z} \in R^{H\times W\times d^1}$ are provided, $\text{DWConv}_{5\times 5,e=1}$ is utilized to capture low-scale features. Next, we need to divide the low-scale features into components $\boldsymbol{Z}_1 \in R^{H\times W\times(\beta_1 d^1)}$, $\boldsymbol{Z}_2 \in R^{H\times W\times(\beta_2 d^1)}$, and $\boldsymbol{Z}_3 \in R^{H\times W\times(\beta_3 d^1)}$ along the channel dimension, where β_1–β_3 denote the rate of division and $\beta_1 + \beta_2 + \beta_3 = 1$. Then, we applied $\text{DWConv}_{5\times 5,e=2}$ and $\text{DWConv}_{7\times 7,e=3}$ correspondingly to $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_3$ to extract medium- and high-scale features $\boldsymbol{Y}_1 \in R^{H\times W\times(\beta_1 d^1)}$ and $\boldsymbol{Y}_2 \in R^{H\times W\times(\beta_3 d^1)}$. Finally, $\boldsymbol{Y}_1$, $\boldsymbol{Y}_2$, and $\boldsymbol{Z}_2$ are combined to create a multiscale contextual feature set, $\boldsymbol{Y} \in R^{H\times W\times d^1} = \text{Concat}(\boldsymbol{Y}_1, \boldsymbol{Y}_2, \boldsymbol{Z}_2)$.

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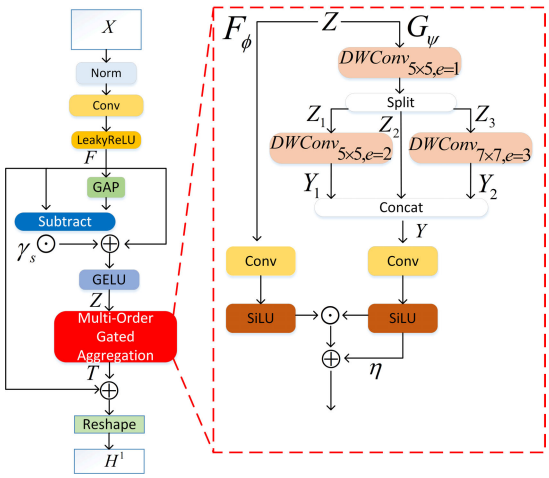


Fig. 6. Structure of the MFAM, $\oplus$ stands for elemental-level addition, and $\odot$ denotes elemental-level multiplication.

To integrate the outputs from various branches, we utilize the SiLU activation function. By taking the output $\boldsymbol{Z}$ of the FD$(\cdot)$ as input, we can represent Moga$(\cdot)$ as follows:

$$\boldsymbol{T} = \text{SiLU}(\text{Conv}(\boldsymbol{Z})) \odot \text{SiLU}(\text{Conv}(\boldsymbol{Y})) \oplus (\eta \cdot \text{SiLU}(\text{Conv}(\boldsymbol{Y}))) \tag{24}$$

where SiLU(Conv($\boldsymbol{Z}$)) denotes the output of the gated branch F_ϕ, while SiLU(Conv(Y)) denotes the output of the contextual aggregation branch G_ψ. We integrate the output of a weighted contextual aggregation branch η·SiLU(Conv($\boldsymbol{Y}$) to enhance the diversity of features, and η is a weighting factor. The features $\boldsymbol{T}$ from Moga$(\cdot)$ and the features $\boldsymbol{F}$ from the CNNs are added at the elemental level to obtain the final output $\boldsymbol{H}^1$, as shown in the following:

$$\boldsymbol{H}^1 = \text{Reshape}(\boldsymbol{T} \oplus \boldsymbol{F}) \tag{25}$$

where Reshape$(\cdot)$ denotes an operation that flattens the spatial dimension ($R^{H\times W\times d^1} \rightarrow R^{HW\times d^1}$). The process of the proposed SSAGU is exhibited in Algorithm 1.

IV. Experiments and Results

This section aims to experimentally verify the effectiveness of our proposed SSAGU. All experiments were conducted using an NVIDIA GeForce RTX 2080Ti GPU (with 22 GB of GPU memory) and i5-12400 CPU, with Python serving as the programming language.

A. Dataset Introduction

We used several widely used hyperspectral image datasets, including the Indian Pines (IP), University of Pavia (UP), due to the utilization of four datasets in this study, it is necessary to include the Houston University dataset here and Salinas datasets to validate the effectiveness of our proposed method.

1) IP Dataset: It was imaged by the Airborne Visual Infrared Imaging Spectrometer (AVIRIS) in 1992 on a patch of IP in Indiana, USA. The dataset has a wavelength range of 0.4–2.5 μm and contains 220 bands, of which 20 bands that could not be reflected by water were excluded. Our study utilized the remaining 200 bands, including 16 land cover categories. The division of training samples, verification samples, and test samples is shown in Table I.

Algorithm 1 Proposed SSAGU Algorithm

Input: Input hyperspectral image $\boldsymbol{X}$; layers of superpixel segmentation l; Initialize variable $i = 1$;
  while $i <= l$ **do**
    if $i = 1$ **then**
      Input $\boldsymbol{X}$ into MFAM to get pixel-level features $\boldsymbol{H}^1$;
    else
      1) Input $\boldsymbol{H}^{i-1}$ into SPS extraction module to get SPS of layer $i-1$: $\{\boldsymbol{sps}_1^{i-1}, \ldots, \boldsymbol{sps}_{m^{i-1}}^{i-1}\}$ and the input feature $\{\boldsymbol{t}_1^i, \ldots, \boldsymbol{t}_{m^i}^i\}$ of the encoder layer i;
      2) Input $\{\boldsymbol{t}_1^i, \ldots, \boldsymbol{t}_{m^i}^i\}$ to $f_{\text{LwGAT}}(\cdot)$ to get superpixel-level feature $\boldsymbol{H}^i$ via Eq.(18), Eq.(19) and Eq.(20);
    end if
    $i++$;
  end while
  $i--$;
  while $1 < i$ **do**
    if $i = l$ **then**
      $\boldsymbol{H}_*^i = \boldsymbol{H}^i$;
    else
      1) Input features $\{\boldsymbol{t}_{*1}^i, \ldots, \boldsymbol{t}_{*m^i}^i\}$ of the decoder layer i are obtained via Eq.(13);
      2) Input $\{\boldsymbol{t}_{*1}^i, \ldots, \boldsymbol{t}_{*m^i}^i\}$ to $f_{\text{LwGAT}}(\cdot)$ to get superpixel-level feature $\boldsymbol{H}_*^i$ via Eq.(18), Eq.(19) and Eq.(20);
    end if
    $i--$;
  end while
  Pixel-level feature $\boldsymbol{H}_*^1$ is obtained via Eq.(13) and Eq.(15);
  Classification result feature $\boldsymbol{P}$ is obtained via Eq.(16);
Output: $\boldsymbol{P}$

TABLE I
Number of Training, Validation, and Testing Samples for the IP Dataset

No	Calss	Train	Validation	Test
1	Alfalfa	2	2	50
2	Corn-notill	29	29	1376
3	Corn-mintill	17	17	800
4	Corn	5	5	224
5	Pasture	10	10	477
6	Trees/Grass	15	15	717
7	Pasture-mowed	1	1	24
8	Hay-windro	10	10	469
9	Oats	1	1	18
10	Soybeans-notil	20	20	928
11	Soybean-mintill	50	50	2368
12	Soybean-cleantill	12	12	590
13	Wheat	5	5	202
14	Woods	26	26	1242
15	Building-Grass	8	8	364
16	Stone-steelTowers	2	2	91
Total		213	213	9940

2) UP Dataset: It is part of the hyperspectral data from images of the city of Pavia, Italy, captured by the German airborne Reflective Optics Spectrographic Imaging System (ROSIS-03) in 2003. The imaging system captured 115 bands in the wavelength range of 0.43–0.86 μm. Twelve of these

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TABLE II
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE UP DATASET

No	Calss	Train	Validation	Test
1	Asphalt	14	14	6603
2	Meadows	38	38	18573
3	Gravel	5	5	2089
4	Trees/Grass	7	7	3050
5	Metalsheets	3	3	1339
6	BareSoil	11	11	5007
7	Bitumen	3	3	1324
8	Bricks	8	8	3666
9	Shadows	2	2	943
Total		91	91	42594

TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE SALINAS DATASET

No	Calss	Train	Validation	Test
1	Brocoli_green_weeds _1	5	5	1999
2	Brocoli_green_weeds _2	8	8	3710
3	Fallow	4	4	1968
4	Fallow_rough_plow	3	3	1388
5	Fallow_smooth	6	6	2666
6	Stubble	8	8	3943
7	Celery	8	8	3563
8	Grapes_untrained	23	23	11225
9	Soil_vinyard_develop	13	13	6177
10	Corn_senesced_green_weeds	7	7	3264
11	Lettuce_romaine_4wk	3	3	1062
12	Lettuce_romaine_5wk	4	4	1919
13	Lettuce_romaine_6wk	2	2	912
14	Lettuce_romaine_7wk	3	3	1064
15	Vinyard_untrained	15	15	7238
16	Vinyard_vertical_trellis	4	4	1799
Total		116	116	53897

bands were excluded due to noise. Typically, images from the remaining 103 bands are utilized. In our study, the image size was 610 × 340, with 42 776 marked pixels containing a total of nine land cover categories, such as Asphalt, Meadows, and Bare Soil. The division of the training samples, verification samples, and test samples is shown in Table II.

3) Salinas Dataset: It was captured by the hyperspectral AVIRIS sensor in the Salinas Valley, California. These data have a spatial resolution of 3.7 m and a size of 512 × 217. The initial data included 224 bands, of which 20 bands with significant absorption of water vapor were removed. The dataset includes 54 129 pixels and comprises 16 land cover categories. The division of training samples, verification samples, and test samples is shown in Table III.

4) Houston University Dataset: The Houston University (HU) dataset provides information about an urban area near the University of Houston campus in USA. The data were collected by the National Center for Airborne Laser Mapping (NCALM) and were used for the 2013 IEEE GRSS Data Fusion Contest. The dataset has a spatial resolution of 2.5 m, with a size of 349 × 1905 pixels. It consists of 144 bands within the range of 364–1046 nm. The study area contains 15 land cover categories. The division of training samples, verification samples, and test samples is shown in Table IV.

TABLE IV
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE HU DATASET

No	Calss	Train	Validation	Test
1	Grass_healthy	26	26	1199
2	Grass_stressed	26	26	1202
3	Grass_synthetic	14	14	669
4	Tress	25	25	1194
5	Soil	25	25	1192
6	Water	7	7	311
7	Residential	26	26	1216
8	Commercial	25	25	1194
9	Road	26	26	1200
10	Highway	25	25	1177
11	Railway	25	25	1185
12	Parking_lot_1	25	25	1183
13	Parking_lot_2	10	10	449
14	Tennis_court	9	9	410
15	Running_track	14	14	632
Total		308	308	14413

B. Comparison Algorithm

The study selected several representative methods for comparison, comprising four CNN-based methods: CNN [47], SSRN [15], LS^{2}CM [16], and HSI-Mixer [48]; and three GNN-based methods: CEGCN [32], WFCG [33], and MSSGU [35]. A division of the training, validation, and testing samples for all comparative methods can be accessed in Tables I–IV.

C. Experimental Setup

We employed the Adam optimizer to train the complete network using a learning rate of 0.0005 and a maximum of 300 iterations. In evaluating the models, we utilized three indicators, namely, average accuracy (AA), overall accuracy (OA), and Kappa coefficient, in order to scientifically assess the performance of the various methods. The network structure of the complete model can be found in Table V, while the specific experimental results are shown in Tables VI–IX. Each result is an average of ten trials, and the optimal values are highlighted in bold.

D. Experimental Results

1) Experimental Results on IP Dataset: The experimental results of our proposed method and various baseline methods on the IP dataset are shown in Table VI. Compared to the baseline methods, our proposed SSAGU delivers optimal values in eight out of 16 categories, showcasing its exceptional performance. Our proposed SSAGU demonstrated the highest classification accuracy regarding OA, AA, and Kappa coefficients. SSAGU showed positive results compared to MSSGU, achieving improvements of 1.36% and 3.29% for the OA and AA, respectively, in addition to a 1.57 increase in the Kappa coefficient. In addition, SSAGU demonstrated a 1.43% increase in OA compared to the WFCG. One reason for this is that SSAGU improves the spectral feature extraction capabilities of graph convolution and addresses the limitation of CNNs in utilizing spatial contextual information. While WFCG understands the significance of spectral and spatial information in feature extraction, it emphasizes solely the

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TABLE V
NETWORK STRUCTURE OF THE PROPOSED METHOD

	Operation		Output Shape	Convsize
	input		(H,W,B)	——
MFAM	$Conv(\cdot)$		(H,W,128)	128(1,1)
	$FD(\cdot)$		(H,W,128)	——
	F_ϕ		(H,W,128)	128(1,1)
	G_ψ	$DWConv_{5\times5,e=1}$	(H,W,128)	(5,5)
		divide	(H,W,48) (H,W,16) (H,W,64)	——
		$DWConv_{5\times5,e=2}$ $DWConv_{7\times7,e=3}$	(H,W,48) (H,W,16) (H,W,64)	(5,5),(7,7)
		$Concat(\cdot)\&Conv(\cdot)$	(H,W,128)	128(1,1)
	$Reshape(\cdot)$		(HW,128)	——
Graph U-Nets	En	$WM(\cdot)\&f_{LwGAT}(\cdot)$	(2048,64)	64
		$WM(\cdot)\&f_{LwGAT}(\cdot)$	(1024,32)	32
		$WM(\cdot)\&f_{LwGAT}(\cdot)$	(512,16)	16
		$WM(\cdot)\&f_{LwGAT}(\cdot)$	(256,8)	8
	De	gU&$f_{LwGAT}(\cdot)$	(512,16)	16
		gU&$f_{LwGAT}(\cdot)$	(1024,32)	32
		gU&$f_{LwGAT}(\cdot)$	(2048,64)	64
		gU	(HW,64)	——
		$Reshape(\cdot)\&Conv(\cdot)$	(H,W,128)	128(5,5)
		$FCL(\cdot)$	(H,W,C)	——

gU denotes graph unpooling operations
En denotes encoder stage, De denotes decoder stage

extraction of spectral features in CNNs, ignoring it in graph convolution. With Laplace smoothing, graph convolution converges the spectral features of similar pixels for improved classification. Hence, acquiring more precise spectral features is critical for graph convolution.

The visual classification results for the IP dataset are presented in Fig. 7. It is evident from Fig. 7(c)–(e) that CNN-based classification methods, including CNN, SSRN, and LS^{2}CM, respectively, have a significant number of misclassification outcomes. The presence of noisy pixels in multiple large homogeneous regions is the primary cause of the suboptimal classification results. In contrast, the GNN-based classification technique registers fewer noisy pixels in homogeneous regions. The outcome implies that the capacity of graph convolution to build long-range dependencies is essential to extract features from data. SSAGU [Fig. 7(j)] outperforms CEGCN [Fig. 7(g)], WFCG [Fig. 7(h)], and MSSGU [Fig. 7(i)] in terms of the smoothness of classification results and the number of misclassified blocks. In addition, SSAGU [Fig. 7(j)] demonstrates superior classification accuracy even with small sample sizes, such as Stone-steelTowers, Soybean-cleantill, and Pasture-mowed (the relevant area is magnified and indicated with a red box).

2) Experimental Results on UP Dataset: The experimental results of our proposed method and various baseline methods on the UP dataset are shown in Table VII. Our proposed SSAGU delivers the optimal values in six out of nine categories. Compared to MSSGU, SSAGU improves the OA, AA, and Kappa coefficients 1.99%, 3.72%, and 2.63, respectively. Compared to WFCG, the OA, AA, and Kappa coefficients of SSAGU increased by 1.22%, 3.49%, and 1.61, respectively. These results demonstrate the competitiveness of SSAGU.

The visual classification results for the UP dataset are presented in Fig. 8. The CNN's classification visualization results [Fig. 8(c)] exhibited a considerable number of misclassified patches and lacked visual appeal. Although the classification visualization results of SSRN [Fig. 8(d)] and LS^{2}CM [Fig. 8(e)] showed significant visual improvement, they still contained notable classification errors. In contrast, GNN-based methods did not suffer from significant classification errors, but differences in some details were observable. For instance, WFCG [Fig. 8(h)] and MSSGU [Fig. 8(i)] misclassified Asphalt as Bricks, whereas SSAGU [Fig. 8(j)] minimized this classification error. In addition, apart from SSAGU [Fig. 8(j)], CEGCN, WFCG, and MSSGU [Fig. 8(g)–(i), respectively] did not exhibit high classification accuracy on Gravel (the accurate rate is below 90%), with significant noisy pixels on the classification result map (the relevant area is magnified and indicated with a red box). This outcome is also evident in Table VII.

3) Experimental Results on Salinas Dataset: The experimental results of our proposed method and various baseline methods on the Salinas dataset are shown in Table VIII. Our proposed SSAGU achieves the best results in nine out of 16 categories. Although MSSGU has achieved high classification accuracy on the Salinas dataset, our proposed SSAGU shows varying degrees of improvement in all three evaluation metrics.

The visual classification results for the Salinas dataset are presented in Fig. 9. As shown in Fig. 9(c)–(e), CNN-based methods, including CNN, SSRN, and LS^{2}CM, respectively, demonstrate a significant quantity of noise in the two most extensive homogeneous regions (Grapes_untrained and Vinyard_untrained), leading to poor visual classification results. Although CEGCN [Fig. 9(g)] and WFCG [Fig. 9(h)] have somewhat reduced the noise pixels in these two regions, misclassifications still persist. In contrast, SSAGU [Fig. 9(j)] significantly enhances the classification accuracy of these two regions, achieving rates of over 99% and 98%. In addition, SSAGU [Fig. 9(j)] achieved optimal and suboptimal classification accuracy for Fallow, Fallow_rough_plow, and Fallow_smooth (the relevant area is magnified and indicated with a red box).

4) Experimental Results on HU Dataset: The experimental results of our proposed method and various baseline methods on the HU dataset are shown in Table IX. Our proposed SSAGU achieves the best results in seven out of 15 categories. Compared to MSSGU and WFCG, the OA of SSAGU increased by 2% and 0.5%, respectively, and AA of SSAGU increased by 1.85% and 0.64%, respectively, which reveals the effectiveness of our method.

The visual classification results for the HU dataset are presented in Fig. 10. Observing the specific differences between different algorithms is challenging in the HU dataset due to its complexity and diversity of ground object coverage types. However, we can analyze the differences in some details. For instance, when considering other classification algorithms, the accuracy of classifying Parking_lot_2 categories is typically

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TABLE VI
ACCURACY OF CLASSIFICATION FOR THE IP DATASET

ID	CNN	SSRN	LS^{2}CM	HSI-Mixer	CEGCN	WFCG	MSSGU	Proposed (SSAGU)
1	**97.42**	68.17	80.01	31.48	33.00	82.16	92.60	91.80
2	55.84	84.63	84.64	91.49	89.18	92.32	**95.10**	93.26
3	65.58	79.72	86.83	77.46	81.48	90.40	91.93	**94.31**
4	68.62	82.04	77.74	85.90	39.60	**95.09**	83.62	86.16
5	85.64	93.74	**94.83**	88.53	86.56	83.73	85.72	87.63
6	78.62	95.06	90.88	98.66	99.19	99.08	98.00	**99.23**
7	92.17	51.43	67.54	46.15	37.50	93.33	84.17	**94.17**
8	79.58	93.30	96.66	**100**	98.74	99.81	99.74	99.72
9	82.95	30.15	69.85	65.00	19.44	**95.00**	83.89	92.22
10	83.44	81.44	89.41	89.88	86.68	93.82	91.39	**94.95**
11	68.56	85.64	89.53	91.61	94.43	95.20	95.92	**96.72**
12	47.06	83.63	73.24	70.20	70.34	91.55	88.22	**94.41**
13	89.24	91.98	94.43	97.17	**98.47**	94.85	98.02	97.62
14	77.52	95.65	95.33	98.07	99.61	98.39	**99.85**	99.67
15	62.86	83.56	81.27	72.63	73.63	94.97	92.75	**97.45**
16	77.09	87.07	83.65	88.42	75.60	93.96	80.00	**94.29**
OA	65.42	84.53	86.61	89.37	88.66	94.31	94.38	**95.74**
AA	75.76	80.45	84.74	80.79	73.97	93.35	91.31	**94.60**
Kappa	82.41	87.38	84.74	87.88	86.98	93.53	93.58	**95.15**

TABLE VII
ACCURACY OF CLASSIFICATION FOR THE UP DATASET

ID	CNN	SSRN	LS^{2}CM	HSI-Mixer	CEGCN	WFCG	MSSGU	Proposed (SSAGU)
1	46.50	96.60	90.02	95.10	97.97	96.88	96.59	**98.58**
2	69.27	95.24	94.57	**98.38**	97.17	98.11	97.28	97.83
3	40.53	80.51	62.04	91.66	85.61	84.80	82.36	**93.46**
4	43.84	90.94	89.32	91.09	90.22	91.92	93.51	**93.65**
5	87.16	99.87	97.22	**100**	99.93	99.92	99.97	99.98
6	47.92	94.12	91.35	99.11	99.57	99.59	99.68	**99.92**
7	40.33	89.37	76.98	81.80	88.33	99.36	98.42	**100**
8	58.43	80.80	80.66	87.13	97.10	96.91	91.19	**99.71**
9	32.64	**99.25**	95.21	72.33	83.96	79.84	86.30	95.65
OA	60.23	92.47	88.34	95.10	96.02	96.58	95.81	**97.80**
AA	51.85	91.86	86.37	90.73	93.32	94.15	93.92	**97.64**
Kappa	90.02	94.99	84.54	93.52	94.78	95.49	94.47	**97.10**

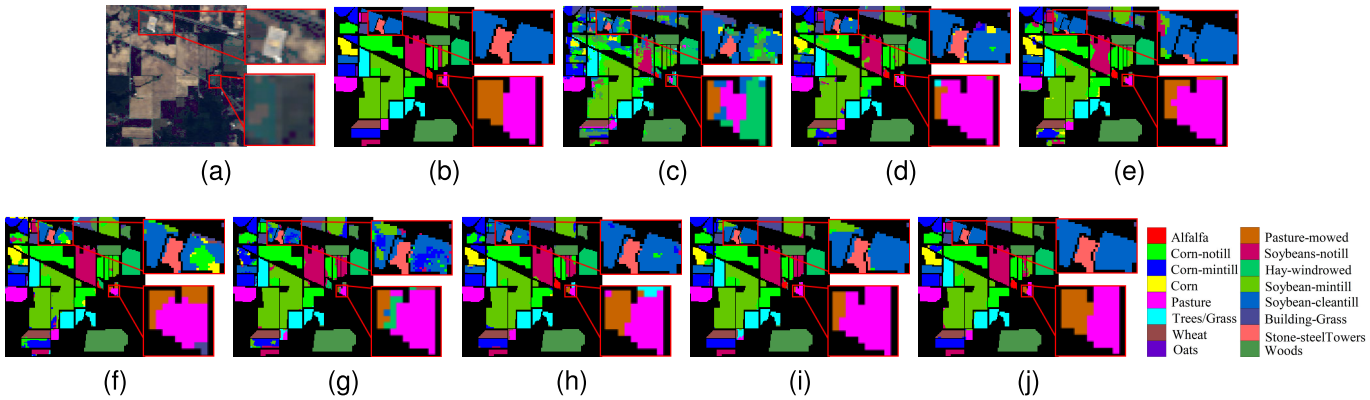


Fig. 7. Visual classification results for the IP dataset. (a) False-color image. (b) Ground-truth map. (c) CNN (OA = 65.42%). (d) SSRN (OA = 84.53%). (e) LS^{2}CM (OA = 86.61%). (f) HSI-Mixer (OA = 89.37%). (g) CEGCN (OA = 88.66%). (h) WFCG (OA = 94.31%). (i) MSSGU (OA = 94.38%). (j) Proposed (SSAGU) (OA = 95.74%).

below 90%. However, for SSAGU [Fig. 10(j)], the accuracy for this specific category can be enhanced to 93.27%. In addition, compared to other categories, the category of Road poses a greater challenge in differentiation. This is because the surrounding areas of the category of Road encompass a multitude of pixels from different categories, resulting in more intricate semantic information. Nevertheless, SSAGU [Fig. 10(j)] can still achieve the highest classification accuracy

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TABLE VIII
ACCURACY OF CLASSIFICATION FOR THE SALINAS DATASET

ID	CNN	SSRN	LS^{2}CM	HSI-Mixer	CEGCN	WFCG	MSSGU	Proposed (SSAGU)
1	81.90	91.16	77.17	**100**	99.31	99.47	97.90	**100**
2	95.57	98.71	92.81	**100**	**100**	**100**	99.99	**100**
3	85.78	94.67	92.78	88.55	93.01	90.94	89.30	**96.32**
4	81.18	96.08	88.76	98.75	97.26	99.18	95.45	**99.53**
5	90.81	94.76	89.07	93.73	97.72	**98.73**	97.27	98.25
6	92.42	99.25	99.60	**100**	99.84	99.98	99.78	99.90
7	86.80	99.21	95.76	94.64	**99.97**	99.92	99.69	99.95
8	80.55	84.90	82.34	83.01	94.00	94.08	99.07	**99.10**
9	90.11	99.59	97.18	**100**	**100**	99.99	**100**	**100**
10	89.68	92.29	89.80	**97.44**	88.69	96.23	93.59	96.89
11	86.93	91.22	87.70	97.00	97.73	99.48	96.90	**99.50**
12	85.37	97.07	89.54	95.74	**100**	99.96	91.96	99.82
13	81.82	92.45	92.76	**100**	94.24	95.43	77.60	98.20
14	85.15	84.41	93.37	**99.81**	98.32	97.60	96.23	97.33
15	72.75	67.79	76.61	90.47	93.84	92.77	96.78	**98.68**
16	77.88	96.48	95.95	80.55	96.83	96.19	89.85	**97.89**
OA	82.84	86.92	85.97	93.84	96.48	96.41	97.09	**99.03**
AA	85.29	92.50	90.08	94.98	96.92	97.50	95.09	**98.83**
Kappa	80.89	85.51	93.16	96.01	96.08	96.00	96.75	**98.92**

TABLE IX
ACCURACY OF CLASSIFICATION FOR THE HU DATASET

ID	CNN	SSRN	LS^{2}CM	HSI-Mixer	CEGCN	WFCG	MSSGU	Proposed (SSAGU)
1	82.04	91.04	88.28	95.37	**95.83**	93.77	93.92	95.12
2	79.11	**97.99**	95.13	82.16	97.15	96.53	88.41	95.10
3	97.84	99.57	97.29	99.09	99.69	**99.99**	99.57	99.94
4	71.30	98.16	97.06	96.48	**99.22**	97.86	98.72	98.90
5	90.89	97.22	97.45	97.68	99.88	99.92	93.26	**100**
6	70.39	97.55	92.34	93.17	93.83	93.06	**99.24**	93.60
7	68.97	89.90	92.57	94.95	**96.78**	96.36	94.65	95.35
8	86.67	**95.98**	95.46	95.07	82.81	82.38	80.54	81.43
9	59.44	88.89	87.99	90.50	91.52	91.53	86.34	**92.58**
10	60.24	86.37	76.01	83.02	98.40	98.12	99.53	**99.60**
11	78.07	92.95	95.23	91.13	95.31	95.16	**98.12**	96.64
12	73.18	91.52	80.59	88.29	91.78	92.22	90.81	**93.20**
13	88.56	88.52	89.72	87.26	73.39	88.31	84.31	**93.27**
14	98.70	98.36	97.64	**100**	99.63	**100**	**100**	**100**
15	93.08	98.41	96.38	98.35	99.97	**100**	99.60	**100**
OA	76.18	93.32	90.40	91.24	94.76	94.83	93.33	**95.33**
AA	79.90	94.16	91.94	92.83	94.35	95.01	93.80	**95.65**
Kappa	74.24	92.77	89.62	90.53	94.33	94.41	92.78	**94.95**

in the category of Road (the relevant area is magnified and indicated with a red box).

E. Parameter Analysis and Ablation Experiments

1) Setting of Hyperparameter r: The parameter r plays a crucial role in the generation of the SPS. To determine the most suitable value for this parameter, we conducted several experiments. We tested the parameter values of 1–6 on four datasets. The results of these experiments are presented in Fig. 11, which displays the OA for each dataset at different values of r. Fig. 11 indicates that the optimal value of r for the IP and Salinas datasets is 2, while for the UP and HU datasets, the optimal value of r is 4. The reason is that the parameter r controls the generation of band weight coefficients in the SPS module. A larger r value will calculate band weights in a more compact representation subspace, requiring fewer parameters. A smaller r value will preserve more original channel features, but a smaller r value will introduce more parameters. We found that IP and Salinas datasets require smaller r values, which may be because IP and Salinas datasets come from the same sensor and have more spectral bands and a longer wavelength range, therefore containing more richer spectral information, requiring more parameter calculations to preserve more original band features. For the subsequent experiments, we will select hyperparameter r with reference to this section.

2) Influence of the Graph Convolution Model: In order to demonstrate that LwGAT does not negatively affect classification accuracy, we utilized GAT and LwGAT as graph

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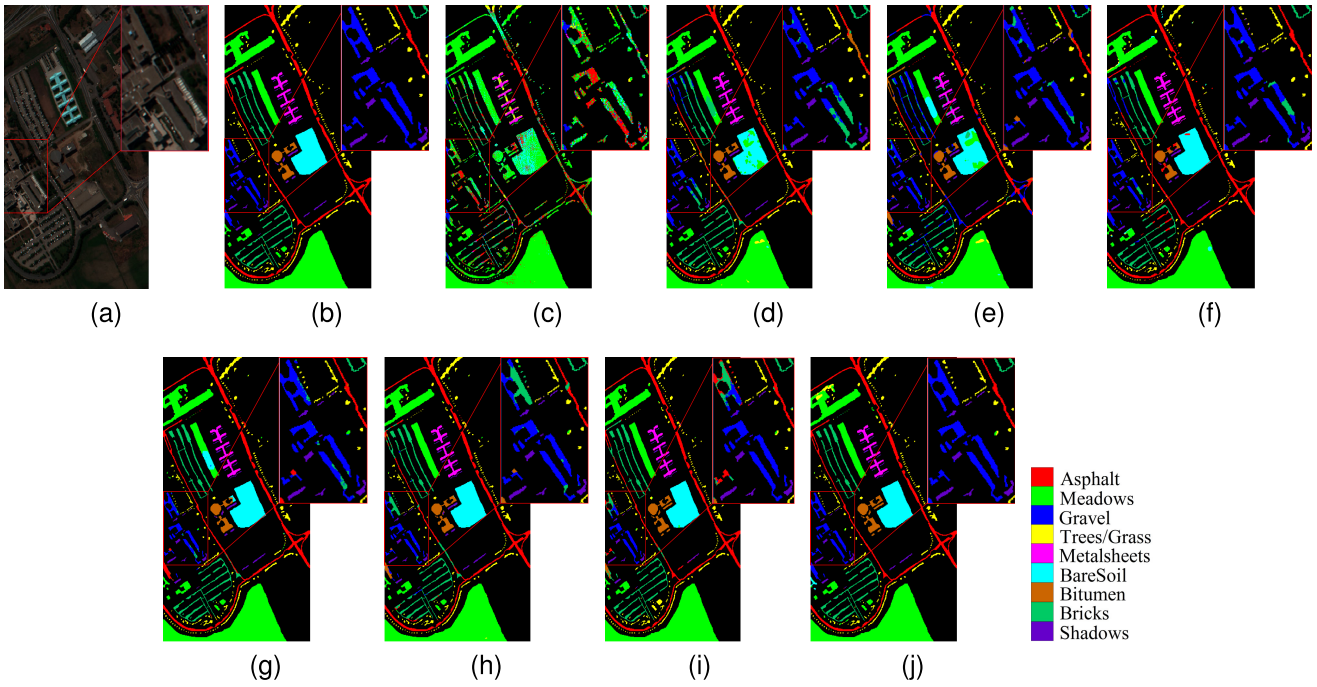


Fig. 8. Visual classification results for the UP dataset. (a) False-color image. (b) Ground-truth map. (c) CNN (OA = 60.23%). (d) SSRN (OA = 92.47%). (e) LS^{2}CM (OA = 88.34%). (f) HSI-Mixer (OA = 95.10%). (g) CEGCN (OA = 96.02%). (h) WFCG (OA = 96.58%). (i) MSSGU (OA = 95.81%). (j) Proposed (SSAGU) (OA = 97.80%).

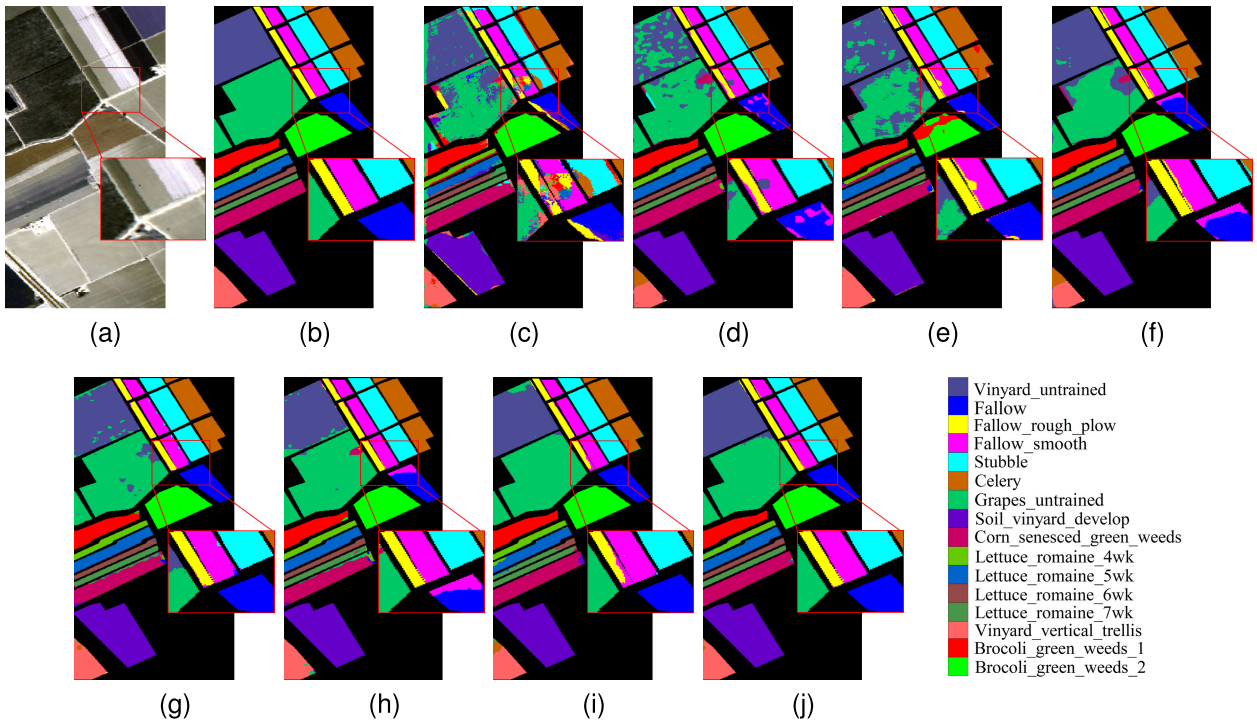


Fig. 9. Visual classification results for the Salinas dataset. (a) False-color image. (b) Ground-truth map. (c) CNN (OA = 82.84%). (d) SSRN (OA = 86.92%). (e) LS^{2}CM (OA = 85.97%). (f) HSI-Mixer (OA = 93.84%). (g) CEGCN (OA = 96.48%). (h) WFCG (OA = 96.41%). (i) MSSGU (OA = 97.09%). (j) Proposed (SSAGU) (OA = 99.03%).

convolution models. In addition, we acknowledged that the selection of neighboring nodes can impact the representation of the central node during graph convolution. Thus, we conducted experiments using one-, two-, and three-hop neighbors as the neighboring nodes of the central node, and the results are presented in Tables X–XIII, where "Time" denotes the time consumption required to construct graph models and perform graph convolution using GAT and LwGAT in the training phase and "parameter" denotes the number of neurons in the feedforward neural network, i.e., $\widetilde{\boldsymbol{a}}^l$ and $\widetilde{\boldsymbol{a}}_1^l$ in (17) and (18). It can be seen from the experimental results that the time consumption required by LwGAT is about 54% of

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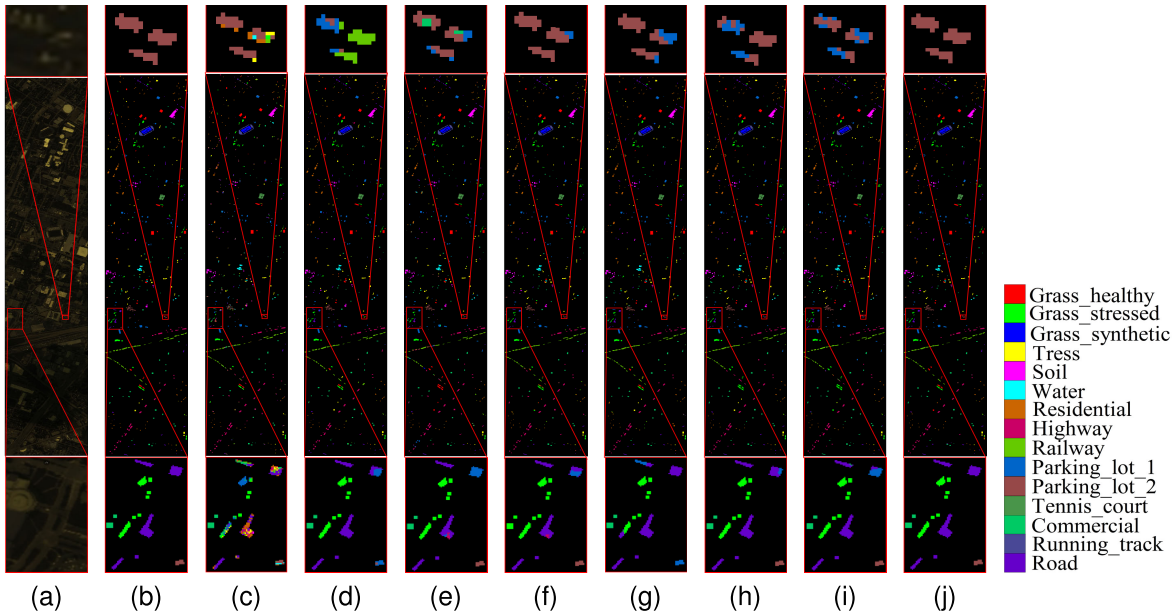


Fig. 10. Visual classification results for the HU dataset. (a) False-color image. (b) Ground-truth map. (c) CNN (OA = 76.18%). (d) SSRN (OA = 93.32%). (e) LS^{2}CM (OA = 90.40%). (f) HSI-Mixer (OA = 91.24%). (g) CEGCN (OA = 94.76%). (h) WFCG (OA = 94.83%). (i) MSSGU (OA = 93.33%). (j) Proposed (SSAGU) (OA = 95.33%).

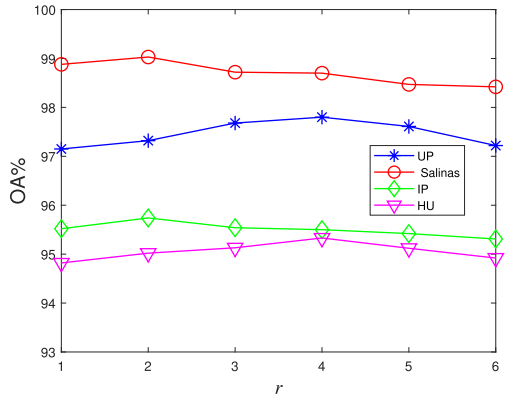


Fig. 11. OA on the four datasets at different r's.

that of GAT on the four datasets. It is worth noting that the reason why the time spent on each dataset is similar is that the number of superpixels obtained when superpixel segmentation is performed on all images is the same (i.e., the number of nodes involved in the construction of graph structure is the same). It is evident that the LwGAT maintains accuracy while reducing parameters and time consumption. In addition, the OA, AA, and Kappa coefficients exhibit a decrease as the number of neighboring nodes increases. This indicates that an increase in neighboring nodes introduces a significant amount of interference information to the central pixel. Specifically, the added neighboring nodes do not belong to the same class as the central node, resulting in the central node aggregating more misinformation and ultimately affecting the final classification results. As a result, our proposed method selects one-hop neighbors of the central node as neighbor nodes.

3) Effects of SPS: The purpose of this section is to demonstrate the usefulness of the extracted SPS through a series of experiments. Specifically, we aim to compare the results of experiments with and without the SPS extraction module added to the model, in order to assess its impact on performance. The experimental results are presented in Table XIV. (NO SPS) refers to directly connecting the features extracted from the encoder stage to the decoder. (SPS) stands for sending the features extracted at the encoder level to the

TABLE X
PERFORMANCE OF GAT AND LWGAT ON THE IP DATASET

	GAT			LwGAT		
	1-hop	2-hop	3-hop	1-hop	2-hop	3-hop
OA	**95.74**	95.59	95.45	**95.74**	95.59	95.45
AA	**94.62**	94.52	94.30	94.60	94.53	94.29
Kappa	95.14	94.95	94.81	**95.15**	94.98	94.81
Time		10.75			**5.82**	
parameters		464			**232**	

TABLE XI
PERFORMANCE OF GAT AND LWGAT ON THE UP DATASET

	GAT			LwGAT		
	1-hop	2-hop	3-hop	1-hop	2-hop	3-hop
OA	97.79	97.71	97.66	**97.80**	97.71	97.64
AA	**97.65**	97.55	97.50	97.64	97.57	97.50
Kappa	97.09	96.93	96.88	**97.10**	96.91	96.89
Time		11.07			**5.95**	
parameters		464			**232**	

TABLE XII
PERFORMANCE OF GAT AND LWGAT ON THE SALINAS DATASET

	GAT			LwGAT		
	1-hop	2-hop	3-hop	1-hop	2-hop	3-hop
OA	**99.04**	98.92	98.85	99.03	98.92	98.85
AA	**98.83**	98.76	98.62	**98.83**	98.78	98.64
Kappa	**99.92**	98.82	98.76	**98.92**	98.81	98.74
Time		11.40			**6.19**	
parameters		464			**232**	

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TABLE XIII
PERFORMANCE OF GAT AND LWGAT ON THE HU DATASET

	GAT			LwGAT		
	1-hop	2-hop	3-hop	1-hop	2-hop	3-hop
OA	**95.34**	95.24	95.18	95.33	95.26	95.17
AA	95.63	95.51	95.44	**95.65**	95.51	95.44
Kappa	94.92	94.88	94.82	**94.95**	94.88	94.80
Time		11.47			**6.22**	
parameters		464			**232**	

SPS extraction module first and then connecting the extracted SPS to the decoder. It is evident that the OA, AA, and Kappa coefficients of the four datasets improve to varying degrees with the inclusion of the SPS extraction module. This suggests that the introduction of the SPS extraction module reduces the redundancy of spectral bands, enabling a more differentiated spectral feature representation of pixels from different classes during feature extraction.

4) Size of Convolutional Kernel: The four datasets mentioned above consist of aerial images. Since the flight plans of the aerial equipment are conducted under clear weather conditions with good visibility and no clouds, the imaging conditions of the aerial images are usually more stable and have higher spatial resolution compared to satellite images. In order to ensure that the size of the convolutional kernel in the MFAM module can simultaneously meet the low spatial resolution of satellite hyperspectral images, we incorporated the Botswana dataset to assess the efficacy of the convolutional kernel size in the MFAM module. The Botswana dataset was obtained from the NASA EO-1 satellite, which captures imagery at a spatial resolution of 30 m. For the Bostwana dataset, 1% of the samples and 1% of the samples are randomly selected from each class as the training set and the verification set, respectively, and the rest are used to evaluate the performance. The MFAM module is composed of three layers of convolutional kernels, each specifically designed to extract features at different scales: low scale, medium scale, and high scale. To visually observe the impact of convolution kernel size on the performance of the MFAM module, we established three groups of convolution kernels with progressively increasing sizes: (3, 3, 5), (5, 5, 7), and (7, 7, 9) (the numbers in parentheses denote the sizes of the kernels used to extract low-, medium-, and high-scale features). The results of these experiments are presented in Table XV.

The result shows that setting the convolution kernel size to (3, 3, 5) results in a limited sensitivity field for each pixel due to the small number of convolution kernel parameters. This limitation leads to the loss of crucial global information. Consequently, the model fails to capture features across the scale space, resulting in poor performance on all five datasets. When the size of the convolution kernel is set to (5, 5, 7), the best performance is achieved on five datasets. Although the performance of the four datasets, which consisted of aerial images, slightly decreased when the size of the convolution kernel increased to (7, 7, 9), these datasets did not exhibit high sensitivity to the increase in convolution kernel size (the decline in accuracy is relatively limited). In contrast, the Botswana dataset experiences a significant decrease in performance. We believe that this is because a larger kernel, while helpful in identifying larger-scale features in the image, results in lower spatial resolution. Since the spatial resolution of the Botswana dataset is low, this means that the level of detail captured in the images is limited. If a large-scale convolutional kernel is applied to these low-resolution images, it will further blur the already limited details. Consequently, the extracted features will become blurry and ambiguous, resulting in the loss of local features. Considering the balance between spatial resolution and contextual information, we set the size of the three-layer convolutional kernel to (5, 5, 7).

5) Verification of Multiscale Features: Features at different scales have varying meanings. Features at a low scale offer higher resolution and more intricate details, but they only encompass local semantic information. Conversely, features at a medium-to-high-scale encompass nonlocal spatial information, aiding in the identification of larger scale features in an image. However, they possess lower spatial resolution and a diminished ability to perceive details. To better analyze the impact of various scale features on different datasets, we examined the performance of low-, medium-, and high-scale features individually, as well as the combined use of multiscale features across different datasets. The specific results can be found in Table XVI. MFAM (low scale) means that we only capture low-scale features (without extracting medium- and high-scale features); MFAM (medium scale) means that after capturing low-scale features, we input all data (without splitting along the channel) into the medium-scale feature extraction module to further extract medium-scale features; and MFAM (high scale) means that after capturing low-scale features, we input all data (without splitting along the channel) into the high-scale feature extraction module to further extract high-scale features. MFAM means that we use features at three scales simultaneously.

The result shows that different datasets necessitate varying scales of features. For instance, the HU dataset exhibits superior performance with medium-scale features, as opposed to low- and high-scale features. Conversely, the Salinas dataset performs better with high-scale features than low- and medium-scale ones. This disparity arises from the Salinas dataset's continuous land cover, which comprises larger homogeneous areas, thereby necessitating large-scale features. The UP dataset and the IP dataset do not yield satisfactory results when employing low-, medium-, or high-scale features individually, and satisfactory outcomes can only be achieved when multiscale features are combined. This is because, although the UP dataset and the IP dataset encompass large homogeneous areas, they also encompass fragmented land cover. Consequently, a more diverse range of features is required to accommodate different land cover types. These findings underscore that single-scale features are inadequate for meeting the requirements of all datasets. The MFAM module, by amalgamating features from different scales, not only fulfills the needs of diverse datasets but also compensates for the limitations of single-scale features, thereby enhancing the performance.

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TABLE XIV
EFFECT OF SPS ON FOUR DATASETS

	HU			Salinas			UP			IP		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
NO SPS	94.58	94.87	94.14	98.29	97.78	98.12	96.56	96.24	95.73	94.68	93.61	94.06
SPS	**95.33**	**95.65**	**94.95**	**99.03**	**98.83**	**98.92**	**97.80**	**97.64**	**97.10**	**95.74**	**94.60**	**95.15**

TABLE XV
EFFECT OF CONVOLUTIONAL KERNEL SIZE ON FIVE DATASETS

	HU			Salinas			Botswana			UP			IP		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
(3,3,5)	94.98	95.25	94.57	98.39	98.13	98.21	93.01	92.36	92.43	97.29	97.22	96.43	95.37	93.93	94.81
(5,5,7)	**95.33**	**95.65**	**94.95**	**99.03**	**98.83**	**98.92**	**94.54**	**94.27**	**94.09**	**97.80**	**97.64**	**97.10**	**95.74**	**94.60**	**95.15**
(7,7,9)	95.12	95.41	94.76	98.95	98.63	98.83	93.80	93.27	93.28	97.62	97.35	96.86	95.56	94.51	94.93

TABLE XVI
PERFORMANCE OF FEATURES AT DIFFERENT SCALES ON FOUR DATASETS

	HU			Salinas			UP			IP		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
MFAM(low-scale)	94.65	94.89	94.20	98.28	97.94	98.13	96.76	96.69	96.26	94.73	93.61	94.06
MFAM(medium-scale)	95.22	95.49	94.83	98.31	97.95	98.11	96.91	96.81	96.57	94.98	93.87	94.28
MFAM(high-scale)	94.72	95.06	94.21	98.84	98.58	98.71	96.85	96.77	96.48	94.79	93.68	94.25
MFAM	**95.33**	**95.65**	**94.95**	**99.03**	**98.83**	**98.92**	**97.80**	**97.64**	**97.10**	**95.74**	**94.60**	**95.15**

TABLE XVII
EFFECT OF β_1–β_3 ON FOUR DATASETS

	UP			Salinas			HU			IP		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
$\beta_1=\frac{1}{3}, \beta_2=\frac{1}{3}, \beta_3=\frac{1}{3}$	97.70	97.71	96.96	98.86	98.63	98.84	**95.35**	**95.70**	**94.97**	95.65	94.71	95.05
$\beta_1=\frac{3}{8}, \beta_2=\frac{1}{8}, \beta_3=\frac{1}{2}$	97.77	97.66	97.06	98.86	98.63	98.84	95.28	95.56	94.90	95.71	**94.76**	**95.15**
$\beta_1=\frac{3}{8}, \beta_2=\frac{1}{2}, \beta_3=\frac{1}{8}$	97.67	97.54	96.91	98.82	98.68	98.82	95.30	95.63	94.93	95.62	94.59	95.10
$\beta_1=\frac{1}{2}, \beta_2=\frac{3}{8}, \beta_3=\frac{1}{8}$	97.61	**97.77**	96.72	98.85	98.62	98.84	95.27	95.58	94.88	95.58	94.41	94.96
$\beta_1=\frac{1}{2}, \beta_2=\frac{1}{8}, \beta_3=\frac{3}{8}$	97.64	97.50	96.86	99.01	**98.84**	98.89	95.25	95.49	94.86	95.61	94.48	95.00
$\beta_1=\frac{1}{8}, \beta_2=\frac{1}{2}, \beta_3=\frac{3}{8}$	**97.80**	97.64	**97.10**	**99.03**	98.83	**98.92**	95.33	95.65	94.95	**95.74**	94.60	**95.15**
$\beta_1=\frac{1}{8}, \beta_2=\frac{3}{8}, \beta_3=\frac{1}{2}$	97.62	97.55	96.85	98.86	98.65	98.73	95.21	95.64	94.82	95.66	94.58	95.05
$\beta_1=\frac{1}{8}, \beta_2=\frac{5}{8}, \beta_3=\frac{1}{4}$	97.58	97.23	96.81	98.70	98.53	98.67	95.01	95.38	94.68	95.50	94.22	94.87
$\beta_1=\frac{1}{8}, \beta_2=\frac{1}{4}, \beta_3=\frac{5}{8}$	97.63	97.45	96.87	98.72	98.59	98.69	95.11	95.35	94.72	95.29	94.27	94.63
$\beta_1=\frac{5}{8}, \beta_2=\frac{1}{4}, \beta_3=\frac{1}{8}$	97.60	97.58	96.90	98.74	98.54	98.71	95.18	95.47	94.78	95.51	94.48	94.94
$\beta_1=\frac{5}{8}, \beta_2=\frac{1}{8}, \beta_3=\frac{1}{4}$	97.71	97.51	96.99	98.84	98.69	98.83	95.29	95.61	94.79	95.53	94.42	94.91
$\beta_1=\frac{1}{4}, \beta_2=\frac{5}{8}, \beta_3=\frac{1}{8}$	97.59	97.56	96.83	98.65	98.58	98.61	95.19	95.54	94.80	95.31	94.27	94.92
$\beta_1=\frac{1}{4}, \beta_2=\frac{1}{8}, \beta_3=\frac{5}{8}$	97.64	97.46	96.92	98.84	98.66	98.82	95.22	95.42	94.84	95.52	94.22	94.89

6) Effects of Hyperparameters β_1–β_3: The purpose of this section is to determine the appropriate values for the hyperparameters β_1–β_3. The above experiments have verified the importance of multiscale features, but we also observed that the importance of features at different scales is different for different datasets. In order to meet the needs of all datasets as much as possible, we try to determine an appropriate segmentation ratio to explore the best combination of features at different scales. Therefore, we considered three sets of value ranges for β_1–β_3: equal proportion segmentation (1:1:1), slight imbalance segmentation (1:3:4), and severe imbalance segmentation (1:2:5). The experimental results are presented in Table XVII. It can be observed from Table XVII that for all datasets, the optimal results are concentrated in equal proportion segmentation and slight imbalance segmentation. This is because the severe unbalanced segmentation is more inclined to use single-scale features, which again verifies the effectiveness of multiscale features. The UP, IP, and Salinas datasets primarily rely on medium- and high-scale features, whereas the HU dataset places equal emphasis on features at all three scales. This is because the UP, IP, and Salinas datasets all contain large-scale homogeneous regions that necessitate broader contextual semantic information. To meet the needs of most datasets, we set the values of parameters β_1–β_3 to $\beta_1 = (1/2)$, $\beta_2 = (1/8)$, and $\beta_3 = (3/8)$.

7) Impact of the Number of Network Layers: The purpose of this section is to analyze how the number of network layers affects the model performance. Lower level encoders and decoders have more detailed superpixel structures that can capture local details, while higher level encoders and decoders have large-scale superpixel structures that can capture large-scale semantic information. Here, "layers" refers to the number of encoders and decoders in the graph U-Nets architecture. To test the performance of the network, we set

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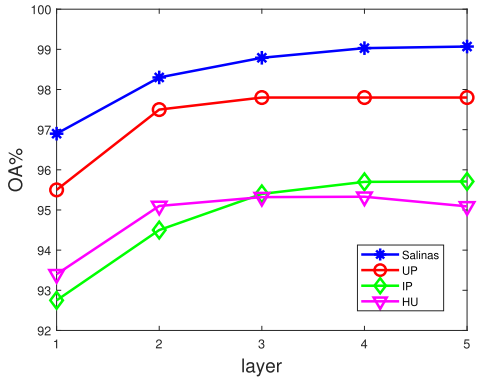


Fig. 12. Effects of different numbers of encoders and decoders on classification performance.

the number of encoders and decoders to 1–5. The number of superpixels in each layer of the encoder and decoder is given as follows: 2048, 1024, 512, 256, and 128. The results of these experiments are presented in Fig. 12. It is clear that the classification accuracy of the IP, Salinas, and UP datasets improves as the number of encoders and decoders increases. Specifically, the classification performance for the IP and Salinas datasets stabilizes when the number of encoders and decoders is increased to 4. The stability of classification performance on the UP dataset is achieved by increasing the number of encoders and decoders to 3. This discrepancy can be attributed to the higher prevalence of discontinuous and fragmented land cover types in the UP dataset compared to the IP and Salinas datasets. Therefore, the UP dataset has less demand for large-scale land cover. For the HU dataset, the classification accuracy reaches a stable state when the number of encoders and decoders is 3, while the performance decreases when the number of codecs is increased to 5. This can be attributed to the fact that most land cover types in the HU dataset are discontinuous. When the number of superpixels is too small, the ground object types covered by these superpixels become too large, which is not suitable for accurately describing the land cover types in the HU dataset. Therefore, we set the number of encoders and decoders to 4.

V. Conclusion

In this article, we proposed a novel HSI classification methodology known as SSAGU. The method integrates spatial and spectral attention using the graph U-Nets architecture for more accurate classification. First, MFAM is employed to process raw hyperspectral data and extract pixel-level features. The module effectively simulates the nonlocal modeling ability of graph convolution by incorporating a multiscale contextual interaction capability; as a result, it enhances the diversity of the extracted pixel-level features. Second, we propose a LwGAT module. This module is capable of reducing the computational resources used when modeling graph structure while ensuring classification accuracy is maintained. Finally, we constructed an SPS extraction module in graph U-Nets. Here, we incorporate the extracted SPS into the LwGAT model to ensure that spectral features extracted from similar pixels are smoother. This also pays attention to Laplace smoothing's practical importance, which is the core of graph convolution. The experimental results on four hyperspectral datasets demonstrate that SSAGU achieves higher performance in terms of evaluation indicators and classification visualization results than other methods.

The future work will concentrate on utilizing deep learning-based pixel segmentation methods to adaptively segment images.

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